

Hackathon Challenge – Project SANKET

A Multimodal AI Assistant for Investigative Interview Analysis (*Sanket – “signal / cue” in Hindi & Sanskrit – reflecting the system’s role as a surfacer of behavioral signals, not a verdict-maker.*)

0. Critical Framing – Read Before Building

This challenge is not a “lie detector.” Decades of peer-reviewed evidence make three things clear:

1. Human and automated deception detection from behavioral cues operates at roughly 54-60 % accuracy in the best meta-analytic findings (Bond & DePaulo 2006; Hartwig & Bond 2014) – barely above chance.
2. Polygraphs and voice-stress analyzers are not admissible in most courts globally and have repeatedly failed independent validation.
3. In India specifically, the Supreme Court in *Selvi v. State of Karnataka (2010) 7 SCC 263* held that involuntary administration of polygraph, narco-analysis, and BEAP tests violates Article 20(3) (protection against self-incrimination) and Article 21 (right to life and personal liberty/privacy). Results from such tests are not admissible as substantive evidence, though information voluntarily disclosed thereafter may be subject to ordinary evidentiary scrutiny.

This system therefore exists to assist a trained officer’s situational awareness during an investigative interview – surfacing observable behavioral signals, voice and language patterns, and physiological indicators that the officer can use to formulate better follow-up questions or to flag segments for senior review. It is explicitly:

- Not evidentiary. Its outputs are inadmissible by design and must be cryptographically tagged as such.
- Not autonomous. The officer retains complete decision authority at every step.
- Not deterministic. Every signal is probabilistic, calibrated, and surfaced with uncertainty.
- Not coercive. The system must not nudge officers toward escalation; it should surface patterns equally consistent with truthful nervousness, cultural difference, fatigue, fear of

police, language disfluency, or neurodivergence.

Every team's submission will be judged in part on how seriously it engages with these constraints (§7, §C of Track E).

1. Background & Context

Investigative interviewing in Indian policing is constrained by:

- Resource asymmetry. Investigating Officers (IOs) frequently conduct dozens of interviews per week with no objective record beyond their own notes. Senior officers reviewing case files have no efficient way to revisit subject behavior across sessions.
- Language complexity. Suspects and witnesses speak Hindi, English, and one or more of 22 scheduled languages, often code-switched within a single sentence. IOs may not share the subject's first language.
- Cultural and procedural drift. Behavioral cues that signal discomfort in one community signal respect or deference in another. Generic Western-trained behavioral models perform poorly on Indian populations.
- Cognitive load. Officers must simultaneously listen, observe, take notes, follow procedure (BNSS 2023, BSA 2023), and plan their next question. Important micro-signals — a sudden change in baseline behavior, a contradiction with a prior statement, a peri-orbital flush coinciding with a specific topic — are routinely missed.
- Lack of structured review tooling. Existing CCTV / interview-room recordings sit on DVRs and are rarely re-watched at depth. Statement analysis, where attempted, is manual and slow.

A growing body of work in affective computing, computer vision, ASR for Indian languages (AI4Bharat, Bhashini, IndicASR), and remote photoplethysmography (rPPG) has matured to the point where a thoughtful, ethically bounded assistive system is technically feasible — provided it is built with calibrated outputs, bias auditing, and human-in-the-loop guarantees from the ground up.

2. The Problem Statement

There is no integrated, multilingual, multimodal assistive platform that helps an Investigating Officer (a) observe a suspect or witness in real time with AI-surfaced behavioral and vocal cues, (b) generate a structured post-session report identifying segments of interest,

inconsistencies, and topic-coupled physiological changes, and (c) correlate observations across multiple sessions or with existing case records — while remaining fully compliant with the Constitutional protections of Article 20(3) and 21, the Selvi (2010) jurisprudence, the Digital Personal Data Protection Act 2023, and the procedural requirements of BNSS 2023.

Existing components — open-source affect recognition (OpenFace, py-feat), eGeMAPS voice features, multilingual ASR (Whisper, IndicConformer, Bhashini), rPPG libraries (pyVHR), pose estimation (MediaPipe) — are individually mature but have never been integrated into a culturally calibrated, legally compliant, officer-facing tool with mandatory uncertainty disclosure and demographic bias testing.

This challenge seeks that integration.

3. Scientific & Ethical Framing (Mandatory Compliance Layer)

Every track and every deliverable must satisfy the following non-negotiable design principles:

Principle	Requirement
Probabilistic outputs only	All cue detections must be expressed as a probability with a confidence interval. No binary “lying / truthful” outputs anywhere in the UI or logs.
Calibrated uncertainty	Each model must report Expected Calibration Error (ECE) ≤ 0.05 on the held-out test set. Miscalibrated models are disqualified from the live mode.
Cue surfacing, not interpretation	The system displays <i>what was observed</i> (e.g., “sustained gaze aversion 03:22-03:31; pitch rose by 38 Hz over baseline”). It does not interpret (“suspect is being deceptive”).
Mandatory baseline window	The system must capture a baseline behavioral profile during a 3-5 minute neutral conversation (name, hometown, weather) before any indexing of “deviations.” All deviation reports are relative to this baseline, not to a population norm.
Demographic fairness	Models must be evaluated for Equal Opportunity Difference and Demographic Parity Difference across (a) sex, (b) age band, (c) skin tone (Fitzpatrick I-VI or Monk Skin Tone scale), (d) primary language. Disparities > 10 percentage points must be reported and mitigated.
Human-in-the-loop	No automated action, alert escalation, or case-record write may occur without an officer-acknowledged click.

Audit logging	Every model output shown to an officer must be logged with input hash, model version, output, and officer’s response. Logs are tamper-evident (hash-chained).
Consent and notification	The interview subject must be informed in their preferred language that AI-assisted observation is in use, and a written consent record (with thumb impression or signed acknowledgement per local procedure) must be captured before the session begins. The system must operate in a consent-absent fallback mode that captures only the officer-facing live notes (no model inference on the subject) if consent is not given.
Inadmissibility tag	Every artifact the system produces is cryptographically signed with an “ASSISTIVE – NOT EVIDENCE” header that survives export.
Discontinuation triggers	If the subject shows signals consistent with a vulnerable status (suspected minor, evident intoxication, signs of acute distress, indications of intellectual disability), the system must surface a procedural advisory and halt cue inference until an officer acknowledges.

A team that builds a brilliant detector but ignores §3 will lose more points than a team that builds a modest detector and treats §3 as a feature. Judges have been instructed accordingly.

4. Challenge Tracks

Track	Focus	Primary Deliverable
T1	Multimodal sensing pipeline (vision, audio, rPPG, optional thermal)	Real-time inference graph on edge hardware, $\leq$ 500 ms cue-to-display latency
T2	Multilingual ASR + linguistic analysis for Indian languages	Speech-to-text + statement analysis pipeline covering $\geq$ 5 languages with code-switch handling
T3	Officer-facing real-time interface	Tablet / dual-screen HUD that surfaces cues without distracting from the interview
T4	Post-hoc deep-analysis console	Reviewable timeline, segment scoring, cross-session correlation, case-record linkage
T5	Ethics, fairness, calibration, and explainability layer	Bias dashboard, calibration report, XAI explanations, audit logging, consent flow

Cross-track integration carries explicit bonus weight. A team that builds a partial slice of every track typically scores higher than a team that perfects one track at the cost of \$3 compliance.

5. Core Technical Objectives

A. Multimodal Sensing & Capture

A.1 Reference capture rig. Interview-room hardware suitable for institutional deployment:

- Multi-angle RGB cameras: 2-3 × Sony IMX-class sensors at 1080p60 minimum (face-on, oblique, wide-room). Global-shutter preferred for clean rPPG. Synchronized via PTP or hardware genlock.
- Beamforming microphone array: 4-mic or 6-mic array (e.g., ReSpeaker, MiniDSP UMA-16) with on-device acoustic echo cancellation. Separate lavalier on the officer for clean speaker separation.
- Optional thermal camera: FLIR Lepton 3.5 or Boson 320 — surfaces periorbital and nasal-region thermal patterns. Mark as *experimental modality* in outputs.
- Ambient sensor: Lux, temperature, humidity — needed to control for environmental confounds in physiological signals.
- Edge compute: NVIDIA Jetson AGX Orin (64 GB) or equivalent — sufficient for real-time multimodal inference with headroom for codec workloads.

A.2 Calibration. A pre-session calibration sequence (color chart, audio sweep, distance markers, rPPG ground-truth via finger pulse oximeter) must be runnable in ≤ 60 seconds and produce a calibration report archived with the session.

A.3 Synchronization. All streams (video, audio, thermal, ambient) must be timestamped on a single monotonic clock with < 50 ms cross-modal skew. Out-of-sync sessions are quarantined automatically.

B. Behavioral Cue Surfacing (Vision)

B.1 Facial analysis.

- Action Unit (AU) detection following the Facial Action Coding System (FACS) — suggested 17 core AUs (AU1, AU2, AU4, AU5, AU6, AU7, AU10, AU12, AU14, AU15, AU17, AU20, AU23, AU25, AU26, AU45). Reference implementations: OpenFace 2.x, py-feat, Libreface, OpenGraphAU.
- Gaze estimation with sub-3° error using L2CS-Net, ETH-XGaze, or MediaPipe Iris. Track gaze direction, fixation duration, saccade frequency, and gaze aversion events relative to officer position.
- Microexpression detection is permitted but must be flagged as low-confidence / contested-science in the UI per Ekman-vs-replication literature debates (Barrett et al. 201U). Use SAMM, CASME II-style architectures if attempted, but no team should anchor its scoring story on microexpressions.
- Head pose & nod/shake detection via 6DoF pose regression.
- Self-touch and grooming gesture detection (hand-to-face, hair touch, mouth cover) via lightweight 2D pose + hand-tracking pipeline.

B.2 Body posture and proxemics.

- Whole-body pose estimation via MediaPipe Pose, YOLOv8-Pose, or RTMPose.
- Postural-shift index (frequency and magnitude), forward/backward lean, shoulder asymmetry.
- Static-baseline-vs-current deviation, expressed as a per-minute scalar in the timeline.

B.3 Critical constraint. Outputs of §B are individual cue scalars — never composite “deception scores.” Composite scoring is permitted only in post-hoc analysis (§F) and only with prominent calibration and bias disclosures.

C. Speech, Voice & Linguistic Analysis (Multilingual)

C.1 Automatic Speech Recognition (ASR).

- Primary stack: AI4Bharat IndicConformer / IndicWhisper, Bhashini APIs (on-prem deployment), or Whisper-large-v3 with Indic fine-tuning. Required language coverage at minimum: Hindi, English, Marathi, Tamil, Telugu, Bengali, Kannada, Malayalam, Punjabi, Gujarati. Bonus for: Urdu, Odia, Assamese, Bhojpuri, regional dialects.
- Code-switch handling: explicit code-switch detection and per-segment language tagging. Hindi-English (“Hinglish”) code-mixing must be handled natively, not by routing to English-only ASR.

- Speaker diarization to separate officer, subject, and any third party (interpreter, lawyer, support person), targeting DER (Diarization Error Rate) $\leq 12\%$ in noisy room conditions.

C.2 Acoustic and prosodic features.

- eGeMAPS feature set (88 features) via openSMILE – F0, jitter, shimmer, harmonics-to-noise ratio, formant bandwidths, voicing-related features. Industry-standard baseline.
- Wav2Vec2 / HuBERT / WavLM embeddings for downstream emotion and arousal classification.
- Speech-rate and pause analysis: words-per-minute, filled-pause rate (“uh”, “umm”, Hindi “matlab”, “haan-na”), unfilled-pause distribution, latency-to-respond on topic shifts.
- Voice stress markers: jitter and shimmer deltas relative to baseline; expressed with explicit uncertainty.
- Critical constraint: Layered voice analyzers (LVA, VSA) and similar proprietary “voice stress” products are prohibited in the POC. They have failed independent validation repeatedly and using them undermines the project’s scientific credibility.

C.3 Linguistic content analysis.

- Statement consistency: cross-segment contradiction detection using sentence-embedding similarity (LaBSE, MuRIL, IndicBERT) and entailment models (XNLI-trained XLM-R).
- Statement-content heuristics drawing on Criteria-Based Content Analysis (CBCA) and Reality Monitoring (RM) literature – sensory-detail density, spatio-temporal anchoring, self-deprecation, unstructured production. These are moderate-evidence cues, not strong ones; their use must reflect that.
- Pronoun and tense shift detection, distancing language, hedging, modal-verb density.
- Named-entity extraction for places, persons, dates, weapons, vehicles – to support cross-referencing with case records (§F.3).
- No use of LIWC-style English-only lexicons for non-English speech without validated translation. Build Indic-language equivalents or abstain.

C.4 On-device processing. Audio and speech processing must run on the edge box. No audio shall leave the agency premises in any form, encrypted or otherwise – this is both a privacy and an evidence-chain requirement.

D. Physiological Inference (Remote)

D.1 Remote photoplethysmography (rPPG).

- Extract pulse signal from RGB face video using POS, CHROM, or learned methods (PhysNet, DeepPhys, EfficientPhys, rPPG-Toolbox baselines).
- Compute heart rate (HR) and heart rate variability (HRV) metrics: RMSSD, SDNN, LF/HF ratio.
- Target accuracy: HR MAE ≤ 5 bpm vs. contact pulse-oximeter ground truth, under typical interview-room lighting (300-500 lux, mixed CRI ≥ 80).
- Surface HRV deviations from baseline as a sympathetic-arousal proxy — expressly not as a deception indicator.

D.2 Optional thermal modality.

- Periorbital and nasal-region thermal patterns sampled at 1-4 Hz.
- Pownall et al. and related work suggest stress-coupled thermal patterns; replication is mixed. Surface as research-grade signal with prominent uncertainty.

D.3 Critical constraint. Physiological cues correlate with arousal — which can be caused by deception, fear, anger, embarrassment, caffeine, pain, exertion, fever, or simply being interrogated by police. The system must display physiological deviations as “elevated arousal” never “stress response to question.” Officers must be trained on this distinction; the documentation must spell it out.

E. Multimodal Fusion & Real-Time Officer Interface

E.1 Fusion architecture.

- Late fusion (per-modality scores combined at decision time) is recommended over early fusion for interpretability — each modality’s contribution must remain inspectable.
- Suggested approaches: gated multimodal units (GMU), attention-based fusion transformers, or simple Bayesian combination of calibrated per-modality posteriors.
- Models must produce per-modality contributions to any composite indicator displayed in SF.

E.2 Live officer interface.

- Discreet dual-screen layout: the subject-facing camera feed on the officer’s primary display; a secondary tablet (or dual monitor) for AI cues, never visible to the subject.

- Cue presentation: rolling 60-second timeline with per-modality lanes (face, voice, posture, physiology). Recent deviations from baseline highlighted; tap-to-expand reveals the underlying measurement.
- Suggested follow-up questions panel (optional): generated by a local LLM (Llama-3-8B, Gemma-3-UB, or IndicBERT-derived) trained or prompted on investigative interviewing best practice (PEACE framework, cognitive interview techniques) — must produce non-leading, open-ended questions and must be reviewable in audit logs.
- No alarms, no buzzers, no haptics. The system surfaces information; it does not signal “lie.”
- Latency budget: cue-to-display ≤ 500 ms end-to-end.

E.3 Discreet earpiece mode (stretch).

- Optional bone-conduction earpiece for the officer with a strict rate-limit (≤ 1 advisory per U0 seconds) and a privacy-mode kill-switch. Subject-audible cues are forbidden.

F. Post-hoc Deep Analysis Console

F.1 Session timeline. Synchronous playback of video, audio, ASR transcript (with per-utterance language tagging), and all per-modality cue tracks. Officer can scrub, bookmark, annotate, and export segments.

F.2 Topic-segmented analysis. Automatic segmentation of the session by topic (using LDA, BERTopic, or rule-based prompts) — physiological and behavioral deviations are then reported *per topic*, which is more interpretable than raw timestamps. (“HR rose 14 bpm above baseline during the 4-minute segment about the vehicle’s whereabouts on 12 March.”)

F.3 Cross-reference layer.

- Named entities from the transcript are matched against CCTNS (Crime and Criminal Tracking Network & Systems) records, prior case files, and existing FIRs through a strictly audited read-only API. No write-back from this system to CCTNS.
- The subject’s prior statements (across this and any prior session in the case file) are loaded as a corpus; the post-hoc tool surfaces contradictions and corroborations with citations.

F.4 Report generation. A structured PDF report per session, signed with the inadmissibility header (§3), containing:

- Session metadata (officer ID, location, consent record hash, model versions, calibration

snapshot).

- Topic segmentation with per-topic behavioral deltas.
- Top 10 “segments of interest” with timestamps and rationale, framed as *suggested for review* – not as findings.
- Demographic-bias disclosure for the models used.
- Calibration plot and ECE for the session.
- Full audit-log hash chain.

F.5 Cross-session correlation. When multiple sessions exist for the same subject, the post-hoc console surfaces longitudinal behavioral baselines and detects within-subject deviations across sessions.

G. Bias, Fairness, Explainability (Mandatory)

G.1 Bias auditing. Required test suite:

- Held-out evaluation across at least 6 demographic strata (sex × age band × skin tone).
- Equal Opportunity Difference and Demographic Parity Difference for every classifier.
- Documented disparity ≤ 10 percentage points or explicit mitigation plan.

G.2 Training-data provenance. Datasets used must be documented with origin, consent basis, and demographic distribution. Western-only datasets (CK+, BP4D, AffectNet) are permitted only for pretraining; any classifier shipped in the POC must be fine-tuned or evaluated on Indian-population data. Suggested sources: IIIT-H Indic expression datasets, agency-collected and IRB-equivalent-approved data, or controlled volunteer sessions with explicit consent.

G.3 Explainability.

- Every cue surfaced live must have a one-tap “why” explanation showing the underlying measurement.
- Post-hoc reports must include SHAP-style feature attributions or equivalent for any composite scores.
- Counterfactual explanations (“this segment was flagged because pitch deviation exceeded threshold X; the threshold was set during the baseline at minute 03”) are required for any segment-of-interest entry.

G.4 Drift monitoring. Models must include a monitoring hook that flags when input-

distribution drift exceeds a threshold (e.g., consistent low-light operation, novel age groups) – and downgrades confidence accordingly.

H. Security, Privacy & Legal Compliance

H.1 Data residency. All raw video, audio, and inferred outputs must remain on agency premises. No cloud processing. No vendor telemetry.

H.2 Encryption.

- Data at rest: AES-256-XTS full-disk encryption on the edge box and on the post-hoc analytics server.
- Data in transit (between rig and on-premise server): TLS 1.3 with mutual certificate authentication; rotating session keys.
- Per-session keys derived from a hardware secure element (TPM 2.0 or equivalent) on the edge box.

H.3 Access control. Role-based access with at least four roles: IO, Senior IO, Reviewer, Auditor. Every read of a session must be logged. Two-officer rule for export to portable media.

H.4 Retention and erasure.

- Retention timelines configurable per case classification, defaulting to BNSS 2023 and DPDP Act 2023 minimums.
- Cryptographic erasure support for closed cases (key destruction renders the encrypted artifact unrecoverable).
- The subject's DPDP-mandated rights (access, correction, erasure) are surfaced via a documented procedure – except where overridden by lawful exception under the Act.

H.5 Legal compliance checklist (must be addressed in the architecture document):

- Constitution of India, Article 20(3) – protection against self-incrimination.
- Constitution of India, Article 21 (post-*Puttaswamy*, (2017) 10 SCC 1) – right to privacy as a fundamental right.
- *Selvi v. State of Karnataka* (2010) – consent and inadmissibility doctrine.
- DPDP Act 2023 – data fiduciary obligations, purpose limitation, consent, breach notification.
- Bharatiya Nagarik Suraksha Sanhita (BNSS) 2023 – procedural requirements for

interrogation and recording.

- Bharatiya Sakshya Adhiniyam (BSA) 2023 — electronic-evidence provisions for any artifact retained even as assistive.
- IT Act 2000 / 2008 amendments — Section 43A, Section 72A obligations.
- Indian Telegraph Act 1885 / TRAI directives — to the extent any captured audio implicates intercept rules.

The architecture document must contain a section-by-section mapping from each technical component to the legal authority that governs it.

C. Key Performance Indicators (KPIs)

#	KPI	Target
1	End-to-end cue-to-display latency (live mode)	≤ 500 ms, U5th percentile
2	ASR Word Error Rate (Hindi, clean room audio)	≤ 12 %
3	ASR WER (English-Hindi code-switched)	≤ 18 %
4	ASR WER (other supported languages, clean)	≤ 20 %
5	Speaker diarization DER (3 speakers, room conditions)	≤ 12 %
6	Action Unit detection F1 (top 7 AUs, Indian-population eval set)	≥ 0.65
7	Gaze estimation angular error	$\leq 4^\circ$
8	rPPG heart-rate MAE (vs. contact ground truth)	≤ 5 bpm
U	Expected Calibration Error (per shipped classifier)	≤ 0.05
10	Demographic parity difference (per shipped classifier)	≤ 0.10
11	Cross-session contradiction recall (curated test set)	≥ 0.70
12	Officer subjective utility score (5-point Likert, $n \geq 10$ mock sessions)	≥ 3.8
13	Audit-log tamper-evidence (red-team test)	100 % detection of tampering

14	Consent-absent fallback compliance (red-team test)	100 % – no model inference on subject
15	Operational uptime per 4-hour session	≥ UU %

KPIs U and 10 are gating – a submission that fails calibration or fairness thresholds may not claim Track T1 or T5 completion regardless of other scores.

7. Hackathon Deliverables

1. Working prototype demonstrating: live multimodal capture on the reference rig (or an approved substitute), real-time officer interface, multilingual ASR, post-hoc console with at least one full mock-session walkthrough.
2. Mock-session dataset of at least 5 recorded interviews with consenting volunteer subjects across 3+ languages, used to validate KPIs. Datasets must include consent records and demographic breakdown.
3. Bias and calibration report (≤ 10 pages) per §G.
4. Legal compliance mapping (≤ 15 pages) per §H.5.
5. Architecture document (≤ 25 pages) covering model choices, dataset provenance, threat model, deviations from this RFP.
6. Source code in a Git repository with a documented OSS license.
7. Live demo (15 min) with one mock session conducted in front of the judges in a language other than English.
8. Recorded demo video (≤ 7 min).

8. Judging Criteria

Dimension	Weight
§3 compliance – scientific honesty, calibration, fairness, consent, audit	25 %
Functional completeness against §5 objectives	20 %
KPI achievement against §6	15 %
Multilingual robustness (real demo in non-English language)	10 %

Legal compliance mapping quality (§H.5)	10 %
Officer UX – does it actually reduce cognitive load?	10 %
Code quality, documentation, reproducibility	5 %
Innovation beyond the stated scope (stretch goals)	5 %

A team that scores zero on §3 compliance cannot win regardless of other scores. This rule is published in advance to incentivize the right engineering culture.

9. Suggested Tech Stack (non-binding)

- Vision: OpenFace 2.x, py-feat, OpenGraphAU, MediaPipe (Face Mesh, Iris, Pose, Hands), L2CS-Net, RTMPose, YOLOv8.
- Audio: openSMILE (eGeMAPS), Wav2Vec2 / HuBERT / WavLM (Hugging Face), pyannote-audio (diarization), librosa.
- ASR (Indic): AI4Bharat IndicConformer, IndicWhisper, Bhashini (on-prem), Whisper-large-v3 with Indic fine-tunes, NeMo.
- NLP (Indic): MuRIL, IndicBERT, XLM-RoBERTa, LaBSE, IndicTrans2 for translation.
- rPPG: rPPG-Toolbox, pyVHR, PhysNet / EfficientPhys checkpoints.
- Fusion & ML infra: PyTorch, ONNX Runtime, TensorRT for Jetson, Triton Inference Server.
- Calibration & fairness: Temperature scaling, Platt scaling, AIF360, Fairlearn, SHAP, Captum.
- Backend: FastAPI or Go (gRPC); PostgreSQL for case metadata; MinIO (on-prem S3) for media artifacts.
- Frontend: React or Flutter for tablet UI; Electron or native Linux for the analyst console.
- Crypto / TPM: libsodium, OpenSSL 3.x, tpm2-tss.
- Audit logging: hash-chained append-only store (e.g., Trillian-style Merkle log).

10. Stretch Goals

- Real-time interpreter mode: Live translation overlay in the officer's preferred language while preserving original audio for record.
- Body-worn rig: A portable version for field interviews (witness statements at incident scenes) with degraded but functional cue surfacing.
- Inter-officer benchmarking: Aggregate (de-identified, with strict access control) analytics across IOs to surface training opportunities – not surveillance of officers.
- Counter-coercion safeguard: Detect officer-side patterns associated with coercive interviewing and flag for senior review. This protects subjects, officers, and the agency.
- PEACE-framework coaching mode: Post-session coaching for the officer on interview technique, separate from any subject-side cue analysis.
- Sign-language recognition: For deaf and hard-of-hearing subjects – ISL (Indian Sign Language) recognition pilot.
- Federated learning: Privacy-preserving model improvement across agencies without raw-data sharing.

11. Constraints & Explicitly Out-of-Scope

- No “deception detection” claim, anywhere, ever – in the UI, in the documentation, in marketing material, in pitch slides. Violations are disqualifying.
- No polygraph integration, no LVA / VSA voice-stress products, no brain-electrical methods. These are precluded by Selvi (2010) and by the scientific record.
- No use of facial recognition for subject identification within the interview tool itself. (Identification is CCTNS's job, performed before the session begins.)
- No cloud APIs for inference on subject data. Cloud-based ASR (Google, Azure, AWS) is disqualifying for production paths; permitted only for offline pretraining of in-house models.
- No predictive policing integration. The system observes; it does not forecast.
- No use of subject's biometric data (face, voice) to train future models without explicit, separable, revocable consent.
- No deployment without an institutional ethics review (Police Research and Development equivalent, or BPRD-mediated review) – flagged here even though the

hackathon itself operates under volunteer-consent mock sessions.

12. Timeline (indicative)

Phase	Duration	Milestone
Kickoff & team formation	Week 0	Teams registered, tracks chosen, dataset-collection plan submitted
Ethics & dataset plan approval	Week 1	Volunteer protocol and consent forms approved
Design submission	Week 3	Architecture document v1, legal mapping v1
Mid-point checkpoint	Week 5	Working live mode (single language), basic post-hoc console
Multilingual integration	Week 7	At least 3 languages live, code-switch handling working
Bias & calibration freeze	Week 8	KPI U-10 verified, mitigation applied if needed
Field test & demo day	Week U	Live mock interview before judges, full post-hoc walk-through

13. Threat Model & Adversarial Considerations

Solutions must, at minimum, consider and address:

- T1 – Officer over-reliance. The system’s outputs become a self-fulfilling prophecy in the officer’s mind. *Mitigation*: probabilistic-only outputs, calibration disclosure, mandatory training, periodic blind A/B comparison.
- T2 – Demographic miscalibration. The system performs worse on certain populations. *Mitigation*: SG testing, fallback to “low confidence” mode.
- T3 – Subject countermeasures. Subject deliberately produces high-arousal baseline. *Mitigation*: long baselines, multi-modal cross-checks, no high-stakes inferences from any single modality.
- T4 – Insider data exfiltration. Officer attempts to export subject data improperly. *Mitigation*: two-officer rule, audit logs, cryptographic export tagging.

- T5 — Tampering with logs to fabricate a record. *Mitigation*: hash-chained append-only logs, periodic external attestation.
- TC — Misuse beyond stated purpose. Tool is used in non-investigative settings (employment screening, etc.). *Mitigation*: hardware-bound license, agency-issued attestation, refusal-to-run outside agency premises.
- T7 — Legal challenge. A court treats system outputs as evidence despite the inadmissibility tag. *Mitigation*: clear procedural training, prominent in-output disclaimers, agency-level standing order forbidding evidentiary use.
- T8 — Adversarial perturbations (e.g., printed glasses, voice cloners) targeting the models. *Mitigation*: out-of-distribution detection, drift monitoring, multimodal cross-check.

14. Closing Note to Participants

This is not a hackathon to “win at lie detection.” That problem is not winnable with current science. This is a hackathon to build a calibrated, fair, multilingual, legally compliant assistive tool that respects the people on both sides of the interview table and that gives an Indian Investigating Officer something genuinely useful — better recall of what happened in the room, structured review of long sessions, multilingual transcripts, and honest observational support — without ever pretending to be more than that.

Teams that internalize this framing will build tools that get deployed. Teams that don’t, won’t.